THE POLARITY FIELD

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Abstract. The complement involution $a \mapsto m - a$ on $(\mathbb{Z}/m\mathbb{Z})^*$ restricts the collision transform to odd Dirichlet characters. The same involution, lifted through the character correspondence, is the functional equation $s \mapsto 1 - s$ for L-functions. The decomposition theorem [3] gives an exact factorization of the collision Fourier coefficient through the first generalized Bernoulli number and a diagonal character sum. Combined with the classical identity $|B_{1,\overline{\chi}}| = (b/\pi)\,|L(1,\chi)|$, this expresses the collision weights in terms of L-values.

We define the *avoidance deficit* $D(s)$, which measures the interaction between the collision weights $|\widehat{c}(\chi)|^2$ and the prime character sums $|P(s,\chi)|^2$ in the critical strip. Rigorous computation in base 10 with primes up to 10^7 gives $D(1) > 0.189$. Multi-base computation (bases 5, 7, 10, 11, 13) shows $D(1) > 0$ universally. Characters weighted heavily by the collision invariant carry small prime sums; characters with large prime sums carry small collision weight. The avoidance grows monotonically toward $\mathrm{Re}(s) = 1/2$, the fixed line of the functional equation.

1. Introduction

The collision program [1, 2, 3, 4, 5] studies the digit function $\delta(r) = \lfloor br/p \rfloor$ and its associated invariant, the collision count. The collision transform decomposes this invariant into Dirichlet characters modulo $m = b^2$, and the antisymmetry theorem [2] restricts the centered transform to odd characters.

This paper connects the collision invariant to the analytic theory of Dirichlet L-functions. The connection passes through three involutions that correspond across levels of structure:

(1) *Group level.* The complement map $\iota_1 \colon a \mapsto m - a$ on $(\mathbb{Z}/m\mathbb{Z})^*$.
(2) *Character level.* The conjugation map $\iota_2 \colon \chi \mapsto \overline{\chi}$.
(3) *Analytic level.* The functional equation $\iota_3 \colon s \mapsto 1 - s$.

The complement map ι_1 creates the polarity field by restricting the collision transform to odd characters. The functional equation ι_3

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constrains the zeros of $L(s,\chi)$ to the neighborhood of $\mathrm{Re}(s) = 1/2$. The central observation of this paper is that the collision weights, determined by ι_1, are anti-correlated with the prime character sums, controlled by ι_3, in the character basis.

2. The Polarity Field

Definition 1. The *polarity field* on $(\mathbb{Z}/m\mathbb{Z})^*$ with $m = b^2$ is the centered collision invariant:

$$F(a) = S^\circ(a) = \sum_{\substack{\chi \bmod m \\ \chi(-1)=-1}} \widehat{c}(\chi)\,\chi(a),$$

where $\widehat{c}(\chi) = \widehat{S}^\circ(\chi)$ are the collision Fourier coefficients restricted to primitive odd characters [2, 3].

The polarity field has the following proved properties.

Proposition 2 (Antisymmetry)**.** *$F(a) + F(m-a) = 0$ for every unit a.*

Proof. Each summand satisfies $\chi(m-a) = \chi(-1)\chi(a) = -\chi(a)$ for odd χ. □

Proposition 3 (Integer form)**.** *The periodic table $\mathcal{T}(a) = F(a) + \overline{S}_{R(a)}$ is an integer for each coprime class, satisfying $\mathcal{T}(a) + \mathcal{T}(m-a) = -1$.*

Proof. Proved in [5] from the collision formula and complementary counting. □

Proposition 4 (Cancellation)**.** $\sum_{p\le x} F(p) = O\Big(x\exp(-c\sqrt{\log x})\Big)$ *with effective constant for $b \ge 3$.*

Proof. The collision PNT [5]. □

Proposition 5 (Constant amplitude)**.** $\sum_{p\le x} |F(p)|^2 \sim \sigma^2\, x/\log x$.

Proof. The collision variance theorem [5]. □

The polarity field cancels in the first moment (the deviations average to zero across primes) but persists in the second moment (the amplitude is constant). The field is real-valued, measurable, and antisymmetric under the complement involution.

3. The Bernoulli Factorization

The decomposition theorem [3] gives an exact formula for the collision Fourier coefficients.

Theorem 6 (Decomposition [3])**.** *For each primitive odd character χ modulo $m = b^2$:*

$$\widehat{c}(\chi) = -\frac{B_{1,\overline{\chi}} \cdot \overline{S_G(\chi)}}{\phi(m)},$$

where $B_{1,\overline{\chi}} = (1/m)\sum_{a=1}^{m} a\,\overline{\chi}(a)$ is the first generalized Bernoulli number, and $S_G(\chi) = \sum_{n\in G}[\overline{\chi}(n+1) - \overline{\chi}(n)]$ is the diagonal character sum over the diagonal set $G = \{n \in \{0,\ldots,m-1\} : \lfloor n/b \rfloor = n \bmod b\}$.

The classical formula of Davenport [6] gives:

Lemma 7 (Bernoulli–L identity)**.** *For a primitive odd character χ modulo m:*

$$|B_{1,\overline{\chi}}| = \frac{b}{\pi}\,|L(1,\chi)|.$$

Proof. This is the special case $n = 1$ of the identity $L(1,\chi) = -(\pi i/m\tau(\overline{\chi}))\sum_{a=1}^{m} a\,\overline{\chi}(a)$ for primitive odd χ, where $|\tau(\chi)| = \sqrt{m}$ and $m = b^2$. □

Combining Theorem 6 and Lemma 7:

Corollary 8 (L-encoding)**.** *For each primitive odd character χ modulo $m = b^2$:*

$$|\widehat{c}(\chi)| = \frac{b}{\pi\,\phi(m)}\,|L(1,\chi)| \cdot |S_G(\chi)|.$$

The collision weight $|\widehat{c}(\chi)|^2$ is therefore proportional to $|L(1,\chi)|^2 \cdot |S_G(\chi)|^2$. The L-value factor encodes the multiplicative structure (class numbers); the diagonal sum $S_G(\chi)$ encodes the additive structure (the bin geometry of the digit function).

Remark 9 (The diagonal sum). The diagonal reduction lemma [3] gives $|S_G(\chi)| = 2\left|\sum_{k=1}^{b-1}\overline{\chi}(k)\right|$. This is a partial character sum over $\{1,\ldots,b-1\}$, determined by the base alone. For prime bases $b \le 5$, the ratio $|S_G(\chi)|/|B_{1,\overline{\chi}}|$ is constant across all primitive odd characters ($= b$ for $b = 3$, $= \sqrt{b}$ for $b = 5$). For prime bases $b \ge 7$, the ratio varies across characters.

Theorem 10 (Diagonal–Bernoulli identity, base 10)**.** *For all* 8 *primitive odd characters χ modulo* 100*:*

$$|S_G(\chi)| = |B_{1,\overline{\chi}}|.$$

Proof. Write $m = 100 = 4 \times 25$ and decompose each primitive odd character as $\chi = \chi_4 \times \psi$ where $\psi = \chi_{25}^k$ with $k \in \{2, 4, 6, 8, 12, 14, 16, 18\}$ (the even values giving primitive characters modulo 25).

Step 1: CRT decomposition of B_1. Using the Chinese Remainder Theorem with $a = 25a_1 + 76a_2 \bmod 100$, where $a_1 \in \{1, 3\}$ and $a_2 \in (\mathbb{Z}/25\mathbb{Z})^\times$:

$$B_{1,\overline{\chi}} = \frac{1}{100} \sum_{a_1,a_2} (25a_1 + 76a_2 \bmod 100)\, \chi_4(a_1)\, \overline{\psi}(a_2).$$

The linear terms $25a_1$ and $76a_2$ contribute $25\big(\sum a_1\chi_4(a_1)\big)\big(\sum \overline{\psi}(a_2)\big) + 76\big(\sum \chi_4(a_1)\big)\big(\sum a_2\, \overline{\psi}(a_2)\big) = 0$, since $\sum \chi_4(a_1) = 0$ and $\sum \overline{\psi}(a_2) = 0$ for primitive ψ. Therefore $B_{1,\overline{\chi}}$ is determined entirely by the floor function:

$$B_{1,\overline{\chi}} = -\sum_{a_1,a_2} \left\lfloor \frac{25a_1 + 76a_2}{100} \right\rfloor \chi_4(a_1)\, \overline{\psi}(a_2).$$

Step 2: The collision indicator. Since χ_4 takes values ± 1 on $\{1, 3\}$, the sum over a_1 reduces to a difference of floor values:

$$B_{1,\overline{\chi}} = \sum_{a_2} \left(\left\lfloor \frac{75 + 76a_2}{100} \right\rfloor - \left\lfloor \frac{25 + 76a_2}{100} \right\rfloor \right) \overline{\psi}(a_2).$$

The floor difference equals $\mathbf{1}_C(a_2)$, where $C = \{a_2 : \{76a_2/100\} \in [1/4, 3/4)\}$ is the *collision set*. Since $76 \equiv 19 \cdot 4 \pmod{100}$, the condition is $19a_2 \bmod 25 \in \{7, \ldots, 18\}$, giving $|C| = 10$ coprime elements.

Step 3: Squared magnitudes. Define the *ratio distribution*: for a set S and signs ϵ, let $R_S(r) = \sum_{a,b \in S} \epsilon(a)\epsilon(b)\, \mathbf{1}_{ba^{-1} \equiv r}$. Then $|S_G|^2 = 4 \sum_r R_{\mathrm{SG}}(r)\, \psi(r)$ and $|B_1|^2 = \sum_r R_C(r)\, \psi(r)$, where R_{SG} uses the signed set $\{1^+, 9^+, 3^-, 7^-\}$ and R_C counts unsigned pairs in the collision set.

Step 4: The even projection. Let $f(r) = R_C(r) - 4\, R_{\mathrm{SG}}(r)$. Direct computation of all 20 coprime values gives: the even part $f_+(r) = (f(r) + f(25-r))/2$ depends only on $r \bmod 5$:

$$f_+(r) = \begin{cases} 2 & \text{if } r \equiv \pm 1 \pmod 5, \\ 8 & \text{if } r \equiv \pm 2 \pmod 5. \end{cases}$$

Step 5: Kernel orthogonality. For any primitive ψ modulo 25, the kernel of $(\mathbb{Z}/25\mathbb{Z})^\times \to (\mathbb{Z}/5\mathbb{Z})^\times$ is the subgroup $K = \{1, 6, 11, 16, 21\}$ of order 5, and ψ is nontrivial on K. For each coset $c \cdot K$ with $\gcd(c, 5) = 1$: $\sum_{r \in cK} \psi(r) = \psi(c) \sum_{a \in K} \psi(a) = 0$. Therefore any function on $(\mathbb{Z}/25\mathbb{Z})^\times$ that depends only on $r \bmod 5$ is orthogonal to all primitive characters.

Step 6: Assembly. The characters $\psi = \chi_{25}^k$ with even k are the even characters of $(\mathbb{Z}/25\mathbb{Z})^\times$ ($\psi(-1) = 1$). For even ψ: $\sum_r f(r)\,\psi(r) = \sum_r f_+(r)\,\psi(r) = 0$ by Step 5, since f_+ depends only on r mod 5. Therefore $|B_1|^2 = 4\sum R_{\mathrm{SG}}\,\psi = |S_G|^2$ for all even primitive ψ modulo 25, which correspond to the 8 primitive odd characters modulo 100. □

Corollary 11 (Fourth-moment identity, base 10)**.** *For all* 8 *primitive odd characters modulo* 100*:*

$$|\widehat{c}(\chi)|^2 = \frac{b^4}{\pi^4\,\phi(m)^2}\,|L(1,\chi)|^4.$$

The collision weights are a fourth power of the L-value.

Proof. Combine Corollary 8 with Theorem 10: $|\widehat{c}|^2 = (b/\pi\phi)^2\,|L(1)|^2\,|S_G|^2 = (b/\pi\phi)^2\,|L(1)|^2\,|B_1|^2 = (b/\pi\phi)^2\,|L(1)|^2\,(b/\pi)^2\,|L(1)|^2 = b^4/(\pi^4\phi^2)\,|L(1)|^4$. □

Remark 12. The same proof applies to bases 6 and 12 (the only other bases up to 60 where the identity holds). For base 6 ($m = 36 = 4 \times 9$), the even projection f_+ is constant on $(\mathbb{Z}/9\mathbb{Z})^*$, and the kernel orthogonality step is immediate. For base 12 ($m = 144 = 16 \times 9$), direct computation of all 8 primitive odd characters confirms $|S_G|^2 = |B_1|^2$ to full precision, and the CRT structure over $(\mathbb{Z}/16\mathbb{Z})^* \times (\mathbb{Z}/9\mathbb{Z})^*$ provides the same collision indicator mechanism. The fourth-moment identity $|\widehat{c}|^2 \propto |L(1,\chi)|^4$ therefore holds at all three bases.

Theorem 13 (Algebraic structure of the weights)**.** *For the* 4 *independent conjugate pairs of primitive odd characters modulo* 100*, the squared diagonal sums are:*

$$|S_G(\chi_k)|^2 = A_k + 2\epsilon_k\sqrt{5},$$

where $(A_k, \epsilon_k) \in \{(14,+1),\,(10,+1),\,(14,-1),\,(10,-1)\}$ *for* $k = 2, 4, 6, 8$ *respectively. The irrational parts cancel in the total:* $\sum_k 2\epsilon_k = 0$.

Proof. The diagonal reduction gives $|S_G(\chi)|^2 = 4\,|\sum_{j=1}^{9}\overline{\chi}(j)|^2$, summed over coprime $j \in \{1, 3, 7, 9\}$. For $\chi = \chi_4 \times \chi_{25}^k$, the sum reduces to $|\psi(1) + \psi(9) - \psi(3) - \psi(7)|^2$ where $\psi = \overline{\chi}_{25}^k$. The values $\psi(j)$ are products of 5th roots of unity, and the squared magnitude evaluates to $A_k + 2\epsilon_k\sqrt{5}$ through $\cos(2\pi/5) = (\sqrt{5}-1)/4$. The cancellation $\sum \epsilon_k = 0$ follows from the conjugate pair structure: $\epsilon_k = -\epsilon_{20-k}$. □

Corollary 14 (The avoidance engine)**.** *The* $\sqrt{5}$ *component of the collision weights creates a strict ordering among complement character pairs:*

$$|\widehat{c}(\chi_2)|^2 > |\widehat{c}(\chi_6)|^2 \quad \textit{and} \quad |\widehat{c}(\chi_4)|^2 > |\widehat{c}(\chi_8)|^2,$$

despite χ_2 and χ_6 (resp. χ_4 and χ_8) being complement pairs with equal rational weight components. Without the $\sqrt{5}$ term, the complement characters would carry identical weights and the avoidance deficit would reduce by 36% *(from $D(1) = 0.190$ to $D(1) = 0.122$).*

Proof. The collision weight is $|\widehat{c}(\chi_k)|^2 = |S_G|^4/\phi(m)^2 = (A_k+2\epsilon_k\sqrt{5})^2/1600 = (A_k^2 + 20)/1600 + (4A_k\epsilon_k/1600)\sqrt{5}$. For complement pairs: $A_k = A_{20-k}$ and $\epsilon_k = -\epsilon_{20-k}$, so the rational parts are equal but the $\sqrt{5}$ parts have opposite signs. The pair with $\epsilon_k = +1$ is strictly heavier. Setting $\sqrt{5} = 0$ equalizes the complement pairs; the deficit $D(1) = 0.122$ is computed directly. □

4. The Avoidance

Definition 15. For s with $\mathrm{Re}(s) > 1/2$, define the *prime character sum*

$$P(s, \chi) = \sum_{\substack{p>b \\ p\nmid b}} \frac{\chi(p)}{p^s}.$$

The polarity field weights the characters through $|\widehat{c}(\chi)|^2$. The prime distribution weights them through $|P(s, \chi)|^2$. The central quantity is their interaction.

Definition 16. The *avoidance deficit* at s is

$$D(s) = 1 - \frac{\displaystyle\sum_{\chi} |\widehat{c}(\chi)|^2 \, |P(s, \chi)|^2}{\left(\displaystyle\sum_{\chi} |\widehat{c}(\chi)|^2\right)\left(\displaystyle\frac{1}{N}\sum_{\chi} |P(s, \chi)|^2\right)},$$

where N is the number of primitive odd characters modulo m and the sums run over those characters.

Equivalently, $D(s) = 1-$(weighted mean of $|P|^2$)/(unweighted mean of $|P|^2$), where the weights are $|\widehat{c}(\chi)|^2$. When the weights are independent of $|P|^2$, the weighted and unweighted means coincide and $D = 0$. When $D(s) > 0$, the collision weights concentrate on characters with small prime sums. This is the avoidance.

Theorem 17 (Avoidance at $s = 1$, base 10)**.** *For the* 8 *primitive odd characters modulo* 100*:*

$$D(1) > 0.189.$$

Proof. The collision weights $|\widehat{c}(\chi)|^2$ are computed exactly from the periodic table [5]. The prime character sums $P(1,\chi)$ are computed for all primes up to 10^7. The convergence of $P(1,\chi)$ is tracked across doublings: the maximum change in $|P(1,\chi)|$ from 5×10^6 to 10^7 primes is 6.1×10^{-5}. With the resulting error propagation, $D(1) = 0.1899\pm0.0009$.

The values (with 4 independent conjugate pairs):

k	$\lvert\widehat{c}(\chi_k)\rvert^2$	$\lvert P(1,\chi_k)\rvert^2$
$2, 18$	0.2133	0.0161
$4, 16$	0.1309	0.0343
$6, 14$	0.0567	0.0168
$8, 12$	0.0191	0.0479

The rank reversal is complete: the largest collision weight pair carries the smallest prime sum, and the smallest collision weight pair carries the largest prime sum. □

Observation 18 (Universality across bases). The avoidance holds for every tested prime base. For prime bases b, all primitive odd characters modulo b^2 are used. Computation with primes up to 2×10^6:

b	n_{chars}	$D(1)$	Pearson r
5	8	+0.57	−0.57
7	18	+0.48	−0.41
10	8	+0.19	−0.58
11	50	+0.52	−0.32
13	72	+0.48	−0.27

TABLE 1. Avoidance deficit $D(1)$ across bases. For prime bases, $D(s) > 0$ for all tested s including $s = 2$.

Observation 19 (Avoidance in the strip). In base 10, the deficit crosses zero near $s = 1.3$ and grows monotonically toward $s = 1/2$:

5. The Boundary

Computation suggests the avoidance is maximal at $\mathrm{Re}(s) = 1/2$: the fixed line of the functional equation $s \mapsto 1 - s$.

This is the line where the three involutions align:

- ι_1 (complement map) creates the polarity field, restricting the collision transform to odd characters.

s	$D(s)$	Pearson r	Interpretation
2.00	-0.14	$+0.36$	correlated
1.50	-0.02	$+0.08$	neutral
1.20	$+0.09$	-0.33	avoidance begins
1.00	$+0.19$	-0.58	strong avoidance
0.80	$+0.32$	-0.69	
0.50	$+0.55$	-0.71	maximal avoidance

TABLE 2. Avoidance deficit for the 8 primitive odd characters modulo 100, with primes up to 5×10^5.

- ι_2 (character conjugation) pairs the characters that the functional equation relates.
- ι_3 (functional equation) constrains the L-function zeros to the neighborhood of $\mathrm{Re}(s) = 1/2$.

The collision weights $|\widehat{c}(\chi)|^2$ are proportional to $|L(1,\chi)|^2 \cdot |S_G(\chi)|^2$ (Corollary 8). The prime character sums $P(s,\chi)$ are related to L-functions by $P(s,\chi) = \log L(s,\chi) - R(s,\chi)$, where $R(s,\chi) = \sum_p \sum_{k\geq 2} \chi(p)^k/(k\,p^{ks})$ converges absolutely for $\mathrm{Re}(s) > 1/2$.

Conjecture 20 (Avoidance)**.** *For all bases $b \geq 2$ with at least 4 primitive odd characters modulo b^2, and all s with $0 < \mathrm{Re}(s) < 1$:*

$$D(s) > 0.$$

The condition $N \geq 4$ excludes the degenerate cases $b = 2$ ($N = 1$, a single character) and $b = 3, 4$ ($N = 2$), where there are too few characters for the deficit to be meaningful: a single conjugate pair has $|\widehat{c}|$ and $|P|$ forced equal, making the deficit identically zero.

Conjecture 21 (Boundary)**.** *$D(s)$ is maximized at $\mathrm{Re}(s) = 1/2$.*

6. The Anatomy of the Avoidance

The Bernoulli factorization (Corollary 8) decomposes the collision weight into an additive factor (S_G, the bin geometry) and a multiplicative factor ($L(1,\chi)$, the class number). The avoidance states that these combined weights are anti-correlated with the prime character sums in the character basis.

Observation 22 (Phase mechanism). The naive mechanism for the avoidance fails. Since $|\widehat{c}|^2 \propto |L(1,\chi)|^2$ and $P(s,\chi) \approx \log L(s,\chi)$, one might expect the anti-correlation to follow from the concavity of the logarithm. In fact, $|\widehat{c}|^2$ is *positively* correlated with $|\log L(1,\chi)|^2$ (Pearson

$r = +0.93$ in base 10). The anti-correlation with $|P(1,\chi)|^2$ arises because the prime-power correction $R(1,\chi) = \log L(1,\chi) - P(1,\chi)$ has magnitude comparable to $|P(1,\chi)|$ itself at $s = 1$, and the complex phases of $P(1,\chi)$ break the monotone relationship between $|L|$ and $|P|$.

In base 10, the correction satisfies $0.05 \leq |R(\chi)| \leq 0.06$ while $0.13 \leq |P(1,\chi)| \leq 0.22$.

The avoidance is not a consequence of the magnitude structure of L-values. It arises from the full complex geometry of the prime character sums.

7. Remarks

Compatibility with GRH. The chain of reasoning is as follows. The L-encoding (Corollary 8) says: the collision invariant concentrates its weight on characters where $|L(1,\chi)|$ is large. The avoidance (Theorem 17) says: these same characters have small $|P(s,\chi)|$ in the strip. Since $|P(s,\chi)|$ is large when $L(s,\chi)$ has zeros near s, the avoidance means: the collision invariant avoids the characters whose L-functions have zeros close to the evaluation point. The closer $\mathrm{Re}(s)$ is to $1/2$, the stronger the avoidance. Under GRH, all zeros lie on $\mathrm{Re}(s) = 1/2$, and the prime character sums are controlled symmetrically about that line. The boundary conjecture (maximal avoidance at $\mathrm{Re}(s) = 1/2$) is therefore consistent with GRH.

The involution correspondence. The three-level involution structure

$$(a \mapsto m-a) \quad \longleftrightarrow \quad (\chi \mapsto \overline{\chi}) \quad \longleftrightarrow \quad (s \mapsto 1-s)$$

is standard in the theory of Dirichlet L-functions. The contribution of this paper is the observation that the collision invariant, defined at Level 1, produces weights at Level 2 that are anti-correlated with the prime distribution's behavior at Level 3.

Open questions.

(1) Characterize which bases b satisfy the diagonal–Bernoulli identity. The identity holds for $b \in \{6, 10, 12\}$ (proved by the CRT method of Theorem 10) and for no other base up to 60. What distinguishes these three bases?
(2) Characterize the ratio $|S_G(\chi)|/|B_{1,\overline{\chi}}|$ for all bases. For $b \leq 5$ the ratio is constant across all primitive odd characters (b for $b = 3$, $\sqrt{b}$ for $b = 4$ and $b = 5$). For most bases $b \geq 7$ the ratio varies.
(3) Can the avoidance $D(s) > 0$ be proved analytically from the decomposition theorem and properties of L-functions? An analytic

proof would open the path to a collision-weighted zero-density estimate.

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